CONNEXUS black sea tech forum | 2026

NEXT-GEN MARITIME AUTONOMY

Modular, Scalable Autonomy Architecture & USV Platforms



ICT CLUSTER VARNA

BSMEPA BULGARIAN SMALL AND MEDIUM ENTERPRISES PROMOTION AGENCY

ВАРНА

BRAIN++ AI FACTORY

ARC FUND



The Global Compass Forum

Pontic Dynamics

Reliability | Autonomy | Resilience

CEO & Founder
Borislav Teberov

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ACT I: THE CALL TO ADVENTURE

"We stand at the dawn of an epochal change—where computing and AI come upfront to lift the crushing physical loads and mortal perils off the crew. AI is born to protect and solve problems, not to substitute the mariners of the world."

The Mariner's Duty

For centuries, seafarers have braved isolation, severe weather, and mortal peril to keep global trade and coastal security moving.

The Call to Expand

Modern ocean expansion—offshore wind farms, subsea infrastructure, and continuous environmental monitoring—demands exponential mission hours.

The Human Limitation

Seafarer fatigue, severe labor shortages, and high-risk environments create a critical operational bottleneck.

The Vision

We do not build autonomous systems to replace the captain; we build intelligence to protect the mariner and solve the ocean's hardest problems.

Key Insight: *The Hero's Quest: The seafarer has always braved the untamed ocean. Today's quest is not to abandon the shipmaster, but to elevate seamanship with resilient autonomous intelligence.*

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COMPANY & MISSION

Company Background

Pontic Dynamics is an autonomous maritime systems developer specializing in USV control infrastructure.

Reliability

Engineering technical systems explicitly for maritime mission success in demanding marine environments.

Autonomy

Designing intelligent onboard software capable of independent, reliable operations without human intervention.

Resilience

Constructing hardware architectures to perform predictably and maintain integrity in high-stakes environments.

Mission Objective

We provide the "brains"—integrated control infrastructure and modular software—required to run modern unmanned surface vessels.

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ACT II: THE REGULATORY FOG

Surging Commercial Demand vs. International Statutory Voids

The Regulatory Fog

Global autonomous shipping regulations remain fragmented, creating an operational vacuum. Sovereign flag states enforce divergent rules while formal IMO MASS Code frameworks evolve slowly.

Surging Market Demand

Energy transition, hydrographic mapping, and harbor security cannot wait for bureaucratic alignment. Operators face multi-year custom integration cycles and regional approvals.

Respect for Seamanship

Autonomy cannot be a faceless black box that banishes maritime heritage. Human-in-the-Loop (HITL) architecture is mandatory, bridging safety with technology.

The Path Forward

Key Insight: *Technology is ready to answer the ocean's call, but navigating the regulatory fog requires an unwavering commitment to Human-in-the-Loop seamanship.*

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ACT III: THE MENTOR & THE TALISMAN

Forging Tools in the Crew's Image

PD Ship Master Deck

Shore Command & Control HMI—strategic command, waypoint routing, and fleet health.

PD Chief Officer

Onboard AI navigation engine—multi-modal perception, route optimization, collision avoidance.

PD Chief Engineer

Subsystem controller & autopilot—managing engine power, electrical distribution, and watchdogs.

PD Data Safe

Hardened black-box telemetry unit—preserving immutable operational logs for transparency.

The Unhackable Talisman

PD Sextant NG

- **Celestial Polarization:** Measures sky polarization fused with IMU data.
- **Zero RF Footprint:** Passive and non-emitting, maintaining integrity during GNSS loss.

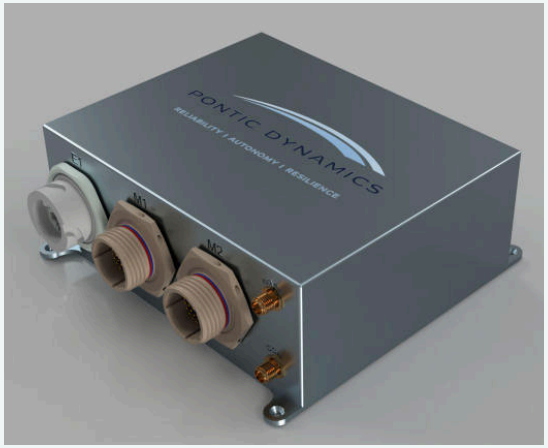
Key Insight

We do not ask the captain to surrender the helm to a nameless machine—we equip the mariner with intelligent, crew-tier extensions.

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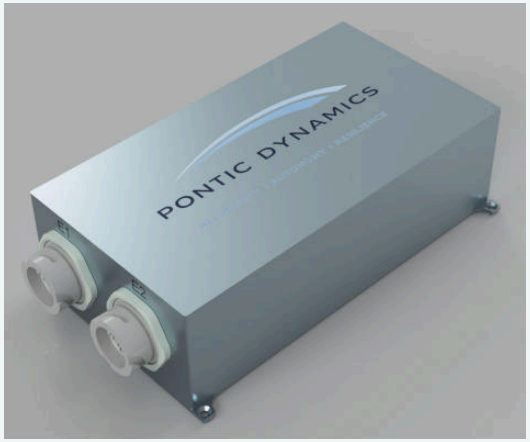
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PRODUCTS



PD Chief Engineer

USV general-purpose subsystem controller and Human-in-the-Loop (HITL) autopilot, engineered to deliver robust marine autonomy alongside reliable remote operator intervention.



PD Data Safe

A ruggedized, secure USV data storage unit engineered for high-capacity mission logging and reliable data preservation in extreme marine environments.



PD Chief Officer

A next-generation navigation system featuring built-in AI data analytics, engineered to provide autonomous decision support and high-precision route optimization for modern marine operations.

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PRODUCTS





PD Ship Master Deck HM

USV Command and Control Software delivers an intuitive, high-reliability interface for autonomous marine operations. Featuring plug-and-play payload integration, advanced waypoint routing, and comprehensive health monitoring, it bridges the gap between shore control and marine autonomy

*Provided picture is for indicative use only and does not represent end product

PD Virtual Cargo Deck

EDGE Core, a hardware-isolated, containerized execution environment that lets developers deploy mission AI applications — SIGINT, ISR, EW, navigation, swarm coordination — directly on autonomous vessels, independent of satellite links..

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ACT IV: CROSSING THE THRESHOLD

Defeating Spoofing, Jamming, and Communication Blackouts

Electronic Warfare & Spoofing

- Littoral waters are subject to false GPS and RF jamming.
- Autonomy must survive signal degradation.

Fortress at the Edge

PD Virtual Cargo Deck

- Hardware-isolated, containerized local AI on the vessel.
- Eliminates heavy coordination with shore operators.

Deterministic Fail-Safe

- Edge watchdogs execute automated Return-to-Base (RTB).
- Safety is guaranteed independent of satellite link.

The Tactical Edge

Key Insight: When external networks fail, the system transitions autonomously to hard-deterministic local protocols to ensure absolute vessel safety.

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ACT V: CONFRONTING THE GUARDIAN

Slaying the Dragon of Marine Insurance

THE ACTUARIAL VOID

Hardened Telemetry

Insurers lack loss history for uncrewed platforms.

PD Data Safe delivers ruggedized, tamper-proof onboard logging to assign liability accurately.

BLACK-BOX HESITATION

Deterministic Control

Underwriters reject unexplainable decision algorithms.

Decoupled safety routines from payload code prove predictable boundaries.

HUMAN ERROR CLAIMS

Automated Safeguards

Human fatigue causes 75–80% of marine casualties.

AI handles continuous watchkeeping and monotonous loads, eradicating traditional H&M claims.

Key Insight

Insurers do not underwrite black-box promises. They insure tamper-proof audit trails, deterministic fail-safes, and systems that protect seafarers from fatigue.

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ACT VI: THE NORTHERN PASSAGE

Why the North Moves Faster

Testbeds & Sandboxes

Maritime authorities in Norway, Denmark, the Netherlands, and the UK established clear autonomous test zones.

Risk Assessment (SORA)

Regulators evaluate risk based on specific operational envelopes rather than waiting for universal type approvals.

Economic & Infrastructure

High density of offshore wind arrays and subsea assets creates urgent demand for uncrewed survey vessels.

Key Insight

Risk-based operational assessments in designated sandboxes prove that we do not have to wait a decade to safely permit uncrewed commercial fleets.

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ACT VII: THE AUGMENTED SEAFARER

Next-Generation Maritime Edge AI & Industry Partnership

The Technical Elixir

Edge Sensor Fusion (`PD Chief Officer`): Onboard processing of polarimetric vision, radar, and thermal optics without latency.

Adaptive COLREGs Compliance: ML algorithms interpreting dynamic multi-vessel encounters in real time.

Hardened Resilience: Architectures built specifically to withstand GPS jamming, spoofing, and harsh sea states.

Call to Fellowship

Collaboration Over Isolation: Building autonomous shipping requires active cooperation between technology creators, regulators, vessel builders, and marine underwriters.

Join the Mission: We welcome discussion, pilot projects, and strategic partnerships across the global maritime industry.

Connect With Us: Follow **Pontic Dynamics** on LinkedIn for operational updates and engineering insights.

Key Insight

Computing and AI are here to solve humanity's hardest problems and protect our seafarers. Together, we chart the future of autonomous shipping.

„Never be afraid to ask questions, because if you never ask, the answer is always no.“

Sheena Yap Chan